DRIVING DEMAND

AN AI-BACKED DRIVER NUDGE SYSTEM FOR DEMAND-AWARE POSITIONING DECISION

TEAM 7

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DRIVER NUDGE SYSTEM

Data-driven insights. Smarter driving. Better outcomes.

THE PROBLEM

Driver Idle Time
Drivers wait in the wrong zones losing earnings while riders wait elsewhere

Zero Visibility
No data or reasoning behind dispatch, drivers can't make informed decisions

Lost Revenue
Supply-demand mismatches mean longer wait times and fewer completed rides

DEMAND INTELLIGENCE PIPELINE

From raw trip data to actionable nudges and validated insights

1 INGEST
Load raw TLC trip data
Apache Spark · Databricks

2 CLEAN
Handle nulls, outliers, and timestamps
Apache Spark · Databricks

3 AGGREGATE
Aggregate trips by ZONE × DAY × DAYPART
Apache Spark · Delta Lake

4 SCORE
Score demand relative to city average
Apache Spark

5 RANK
Select Top-N zones per time window
Apache Spark

6 LLM NUDGE GENERATION
Convert ranked insights into clear, actionable driver messages
OpenAI API · Databricks

7 VALIDATION
Evaluate predictions using March holdout data
Apache Spark · Matplotlib

KEY IMPACT

- Reduce driver idle time by guiding them to high-demand zones
- Increase completed rides and reduce passenger wait times
- Improve driver earnings through data-driven recommendations
- Deliver explainable nudges backed by **holdout validation**

TECH STACK

Apache Spark · Databricks · Delta Lake · OpenAI API · Matplotlib